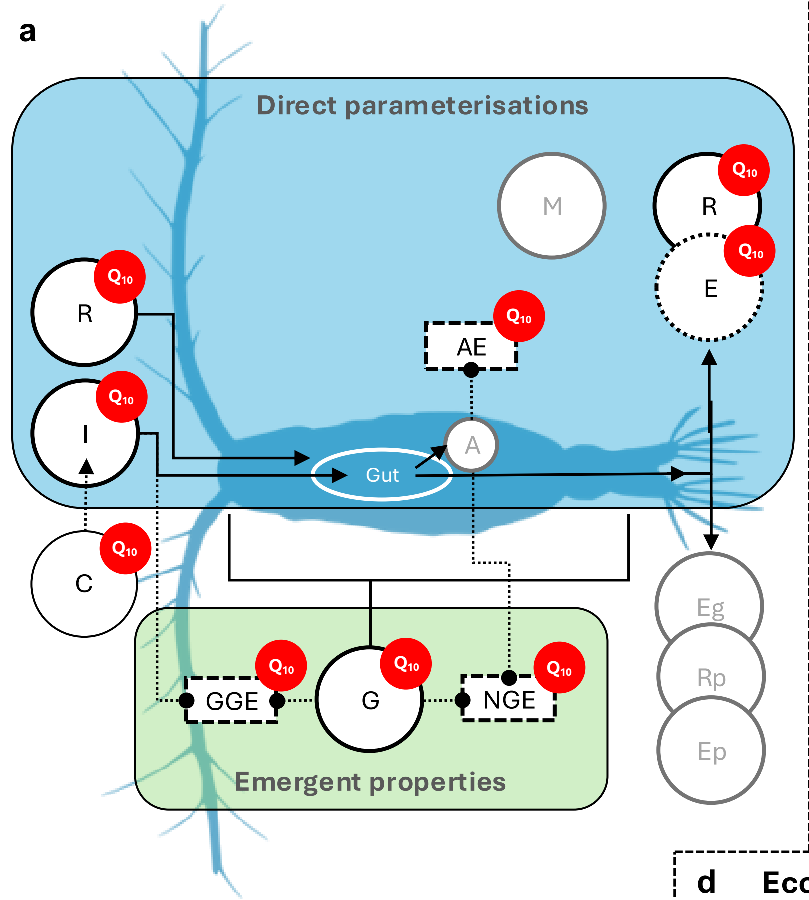
a



b Zooplankton energy budget equation

$$\underbrace{\frac{dZ_c}{dt}}_{\text{Growth}} = \overbrace{AE}^{\text{Assimilation efficiency}} \times \overbrace{IZ_c}^{\text{Ingestion}} - \overbrace{RZ_c}^{\text{Respiration}} - \overbrace{MZ_c^2}^{\text{Predation mortality}}$$

$$\underbrace{\boxed{(-EZ_N)}}_{\text{Excretion}}$$

c Biological rates & model equations

C = Clearance

I = Ingestion = $\dfrac{F(T)_I I_{max} P}{K_{1/2} + P}$

G = Growth = $\dfrac{dZ_c}{dt}$

R = Respiration = $F(T)_R R_{ref}$

E = Excretion = $F(T)_E E_{ref}$

A ≈ G + R = Assimilation

Eg = Egestion

Rp = Reproduction

Ep = Egg production

M = Predation mortality = $F(T)_M M Z_c^2$

d Ecological efficiencies & model equations

AE = Assimilation efficiency = $\dfrac{A}{I} = F(T)_{AE} AE_{ref}$

GGE = Gross growth efficiency = $\dfrac{G}{I} = \dfrac{\left(\frac{dZ_c}{dt}\right)}{IZ_c}$

NGE = Net growth efficiency = $\dfrac{G}{A} = \dfrac{\left(\frac{dZ_c}{dt}\right)}{\frac{dZ_c}{dt} + RZ_c}$